

MAINTENANCE OPERATIONS CLIENT

Maintenance Performance Analysis

Operational Analysis | Maintenance Performance & Management Visibility

Document Control	Detail
Engagement	Aug-Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Confirmed Finding	92% overall completion rate
Related Artifacts	BRD, User Stories, KPI Catalogue, Data Dictionary, Data Cleaning Methodology
Status	Evidence-led portfolio analysis; unsupported historical values are intentionally omitted

Analysis integrity: This document analyzes only findings retained from the engagement. Exact historical team rankings, technician scores, backlog counts, overdue percentages, priority distributions, category frequencies and asset rankings are not retained and are not fabricated.

1. Executive Analysis

The maintenance performance analysis converted approximately 10,000 historical maintenance records across 20 teams into management-level operational visibility. The analysis focused on completion, work status, overdue work, timeliness, priority, team and technician performance, backlog, workload distribution, recurring maintenance categories and equipment-related patterns.

Headline finding: The analysis surfaced an overall maintenance completion rate of 92%.

The 92% result is the strongest retained quantitative KPI from the engagement. It describes the observed completion position within the analyzed scope; it is not evidence that the project improved completion by 92% or increased completion to 92%.

2. Analytical Objective

- Establish an overall view of maintenance completion.
- Separate completed, open and pending work for operational visibility.
- Identify overdue maintenance work using a consistent business rule.
- Understand completion-time performance.
- Compare operational patterns across teams and technicians.
- Assess priority distribution and high-priority pending work.
- Identify backlog and workload imbalance.
- Identify recurring maintenance categories.
- Analyze maintenance patterns associated with assets/equipment.
- Provide the Maintenance Manager with a reusable analytical view in Power BI.

3. Analysis Framework

Analysis Layer	Business Question	Confirmed Evidence
Overall performance	What proportion of assigned maintenance work is complete?	92% completion rate
Work status	What work is completed, open or pending?	These status areas were analyzed
Exceptions	What work is overdue?	Overdue work was identified
Timeliness	How long does work take to complete?	Average completion time was analyzed
Teams	How does performance differ across teams?	Team differences were identified
Technicians	What patterns exist at technician level?	Technician performance was analyzed
Workload	Is work distributed evenly?	Workload imbalance was identified
Priority	What critical work remains pending?	High-priority pending work was identified
Categories	Which maintenance categories recur?	Recurring categories were identified
Assets	What equipment-related patterns exist?	Equipment issues/patterns were identified

4. Overall Completion Performance

Confirmed rule: Completion Rate = Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100

Observed result: Overall Completion Rate = 92%

Completion Rate provided the top-level measure of maintenance execution. Because the numerator and denominator are explicitly defined, the KPI could be used as a stable management measure after status standardization and duplicate control.

Interpretation	What Can Be Concluded
92% completion	A high proportion of eligible assigned maintenance tasks in the analyzed scope were classified as completed.
Remaining work	The completion rate implies that not all eligible assigned work was completed; exact composition/counts are not retained.
Management use	Completion provides an executive indicator but must be read with overdue, backlog, priority and workload views.
Causal limit	The KPI does not establish why work was or was not completed.
Project-impact limit	The project surfaced the KPI; it did not establish a measured improvement to 92%.

5. Work Status Analysis

Completed, open and pending maintenance work were confirmed analytical areas. Status standardization was therefore a prerequisite for meaningful comparison.

Status View	Analytical Purpose	Evidence Boundary
Completed	Understand executed maintenance workload.	Exact historical count not retained.
Open	Understand work remaining open under the operational status logic.	Exact historical count/mapping not retained.
Pending	Understand work categorized as pending.	Exact historical count/mapping not retained.
Status mix	Provide context around the overall completion position.	Exact historical distribution not retained.

Control: Open and Pending must not be treated as interchangeable unless the reconstructed status mapping explicitly supports that meaning.

6. Overdue Performance

Confirmed business rule: Due Date < Current Date AND Status != Completed

Overdue analysis converted due-date and status information into an exception-management view. This was important because an overall completion rate alone cannot show whether remaining work has crossed its due date.

Analytical Check	Purpose
Due Date validity	Prevent missing/invalid dates from creating false overdue records.
Status != Completed	Exclude completed work from overdue classification.
Overdue by team	Support operational ownership analysis where data is valid.
Overdue by priority	Help distinguish critical overdue work from lower-priority work.
Overdue by asset/category	Support pattern investigation where attribution is available.

Retained finding: Overdue work existed in the analyzed maintenance operation. Exact overdue counts and percentages are not retained.

7. Completion-Time Analysis

Average completion time was analyzed to add a timeliness perspective to the completion KPI. The exact historical calculation convention is not retained, including which start/reference date was used in every case.

Analysis Principle	Portfolio Treatment
Eligible population	Use completed records with valid dates required by the rebuilt definition.
Start/reference date	Define explicitly in the portfolio reconstruction before calculating.
Completion date	Use valid Completion Date.
Invalid intervals	Investigate negative/impossible intervals rather than silently accepting them.
Historical result	Do not invent an exact average completion-time value.

Interpretation: Completion Rate answers whether work is completed; Completion Time adds how quickly eligible work is completed. They should be interpreted together.

8. Team Performance Analysis

The analysis identified differences in performance across teams. Team names first required standardization because inconsistent labels could otherwise fragment the same logical team across reporting categories.

Team Analysis	Purpose	Retained Finding / Boundary
Assigned workload	Compare maintenance work volume across teams.	Workload differences could be analyzed; exact volumes not retained.
Completion	Compare completion using the governed completion definition.	Team differences were identified; exact percentages/rankings not retained.
Overdue	Identify team-level exception patterns.	Supported analytically; exact counts not retained.
Completion time	Compare timeliness where valid.	Supported analytically; exact team values not retained.
Workload balance	Assess uneven distribution.	Workload imbalance was identified; exact threshold not retained.

Analytical caution: A lower team completion rate or higher workload does not by itself prove poor team performance. Work mix, priority, asset complexity and other operational context may matter.

9. Technician Performance Analysis

Technician-level analysis provided a more granular operational view. Because the exact historical scoring method is not retained, the portfolio should represent technician performance through a balanced set of operational measures rather than inventing a composite score.

Measure Family	Use
Assigned workload	Understand volume attributed to each technician.
Completed work	Understand executed workload.
Completion rate	Compare completion within valid technician attribution.
Overdue work	Identify exception burden.
Completion time	Assess timeliness where valid dates exist.

Evidence boundary: Technician performance was analyzed, but exact historical rankings, scores and technician names are not retained for public use.

10. Priority Analysis

Priority analysis was used to distinguish maintenance workload by criticality and to surface high-priority work that remained pending.

Priority Lens	Management Value
Work distribution by priority	Understand the composition of maintenance workload.
Pending by priority	Identify unresolved work within each priority group.
High-priority pending	Focus attention on critical work that remains incomplete.
Overdue by priority	Combine lateness and criticality for exception review.

Retained finding: High-priority pending work was identified. Exact counts, percentages and priority labels are not retained.

11. Workload Imbalance Analysis

The analysis identified workload imbalance across the operation. This finding should be treated as a distribution signal, not as automatic evidence of underperformance.

Analytical Lens	Interpretation
Work volume by team	Shows relative assigned workload.
Work volume by technician	Shows relative technician attribution where valid.
Completion alongside workload	Helps avoid judging completion without workload context.
Overdue alongside workload	Helps assess whether exception burden is concentrated.
Priority alongside workload	Helps determine whether workloads differ in criticality.

Historical boundary: The exact imbalance threshold, ratio or percentage used historically is not retained. The portfolio must not invent one.

12. Recurring Maintenance Category Analysis

Recurring maintenance categories were identified through category-level analysis. This provided a way to distinguish repeated types of maintenance activity from isolated records.

Analysis	Purpose
Category frequency	Identify categories appearing repeatedly.
Category over time	Observe recurring patterns where valid dates are available.
Category by team	Understand operational concentration where meaningful.
Category by asset/equipment	Explore whether repeated maintenance activity is associated with particular equipment groups.
Category by status/overdue	Assess whether certain categories carry more outstanding work.

Causal boundary: Recurring categories indicate repeated activity. They do not establish the underlying root cause without additional operational evidence.

13. Asset / Equipment Analysis

Asset/equipment analysis provided visibility into where maintenance activity and issue patterns were concentrated. Equipment-related issues/patterns were a confirmed finding from the engagement.

Asset Lens	Analytical Purpose
Maintenance volume	Understand activity associated with assets/equipment.
Completion	Compare execution by asset/equipment where meaningful.
Overdue	Identify equipment-associated outstanding work.
Category	Understand types of maintenance activity associated with equipment.
Recurring activity	Identify equipment with repeated maintenance records for further investigation.

Interpretation boundary: Repeated maintenance records can flag an equipment area for investigation but do not by themselves prove equipment failure or root cause.

14. Integrated Performance Interpretation

The value of the analysis came from combining measures rather than reading a single KPI in isolation. A 92% completion rate provides the executive position, while overdue, backlog, priority, team, technician, category and asset views explain where management attention may be required.

If Management Sees...	Then Review...	Reason
Strong overall completion	Overdue + high-priority pending	A strong headline KPI can coexist with critical exceptions.
Team differences	Workload + priority mix	Different work volumes/criticality may affect comparison.
High overdue burden	Team + technician + asset/category	Locate operational concentration before acting.
Recurring categories	Asset + location + team context	Understand where repetition is concentrated.
Workload imbalance	Completion + overdue + priority	Separate volume imbalance from execution performance.

15. Confirmed Findings Register

Finding	Evidence Level	Safe Portfolio Statement
92% completion rate	Confirmed quantitative	The dashboard surfaced a 92% overall completion rate.
Backlog	Confirmed analytical finding	Backlog was identified in the maintenance workload.
Overdue work	Confirmed analytical finding	The analysis identified maintenance work that was overdue.
Team differences	Confirmed analytical finding	Performance differences were visible across teams.
Workload imbalance	Confirmed analytical finding	The analysis identified workload imbalance.
High-priority pending work	Confirmed analytical finding	High-priority pending work was surfaced for management visibility.
Recurring categories	Confirmed analytical finding	Recurring maintenance categories were identified.
Equipment issues/patterns	Confirmed analytical finding	Equipment-related patterns/issues were identified for further operational attention.

16. What the Evidence Does Not Support

- An exact historical backlog count or backlog percentage.
- An exact historical overdue count or overdue percentage.
- Exact team completion percentages or team rankings.
- Exact technician scores or rankings.
- Exact priority distributions or high-priority pending counts.
- Exact recurring-category frequencies or rankings.
- Exact asset/equipment rankings or failure rates.
- A specific historical workload-imbalance threshold.
- An exact historical average completion-time value.
- A claim that the project increased completion to 92%.
- A causal claim that a particular team, technician, category or asset caused the observed operational pattern.

17. Recommendations Boundary

Recommendations were made and implemented during the engagement, but their exact historical wording is not retained. This analysis therefore stops at the confirmed findings rather than reverse-engineering recommendations and presenting them as historical facts.

Portfolio rule: Any recommendation created later from these findings must be labeled 'Portfolio-proposed recommendation based on confirmed finding.'

18. Management Reporting Impact

Confirmed efficiency outcome: Approximately 3-4 hours of weekly manual reporting effort were eliminated.

The reporting benefit is separate from maintenance-performance outcomes. It reflects reduced recurring reporting effort after the Power BI solution was introduced. The evidence does not identify the specific individual whose time was saved, so the portfolio should not assign that saving to a named role.

19. Analysis-to-Dashboard Mapping

Analysis Area	Portfolio Dashboard Placement*
Overall completion and work status	Executive Maintenance Overview / Work Order & Completion Analysis
Overdue and completion time	Work Order & Completion Analysis
Team and technician performance	Team & Technician Performance
Workload imbalance	Team & Technician Performance
Backlog and priority	Backlog, Priority & Asset Analysis
Recurring categories	Backlog, Priority & Asset Analysis
Asset/equipment patterns	Backlog, Priority & Asset Analysis

*** Reconstruction note:** Dashboard page names are proposed portfolio architecture. The exact historical page names/layout are not retained.

20. Analytical Validation Framework

Validation	Expected Control
Completion	Reconcile Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100.
92% presentation	Label as observed overall completion rate, not project improvement.
Overdue	Every overdue record must satisfy Due Date < Current Date AND Status != Completed.
Team analysis	Use standardized team labels.
Technician analysis	Handle missing technician attribution transparently.
Completion time	Use valid dates and documented rebuilt calculation convention.
Priority	Use standardized supported values; retain unknowns.
Backlog	Use one explicit reconstructed definition consistently.
Categories	Standardize labels before frequency/trend analysis.
Assets	Avoid unsupported causal interpretation.

21. Traceability to Requirements

Analysis Area	BRD	User Story	KPI Catalogue
Completion	BR-02, BR-03	US-01, US-02	KPI-01, KPI-02
Status	BR-03 to BR-05	US-02	KPI-02 to KPI-04
Overdue	BR-06	US-03	KPI-05
Completion time	BR-07	US-04	KPI-06
Priority	BR-08, BR-13	US-05	KPI-07, KPI-11
Team	BR-09, BR-12	US-06	KPI-08, KPI-12
Technician	BR-10	US-07	KPI-09
Backlog/workload	BR-11, BR-12	US-08	KPI-10, KPI-12
Category/assets	BR-14, BR-15	US-09	KPI-13, KPI-14

22. Performance Analysis Summary

Analytical story: 92% overall completion -> examine remaining work -> identify overdue/backlog/priority exceptions -> compare team and technician patterns -> examine workload/category/asset concentration -> provide management visibility through Power BI.

This document is the analytical interpretation layer of the case study. The next operational-analysis artifact should deepen the exception-management story through Backlog & Overdue Work Order Analysis without inventing unavailable historical values.